18	0.194	0.391	1.608
19	0.187	0.403	1.597
20	0.180	0.415	1.585
21	0.173	0.425	1.575
22	0.167	0.434	1.566
23	0.162	0.443	1.557
24	0.157	0.451	1.548
25	0.153	0.459	1.541

Table 9.23

Summary

In this chapter we have acquired the knowledge of

- **Method of Moving Averages**

  Three year moving averages: $\frac{a+b+c}{3}, \frac{b+c+d}{3}, \frac{c+d+e}{3}, \frac{d+e+f}{3}, \ldots$

  Four year moving averages: $\frac{a+b+c+d}{4}, \frac{b+c+d+e}{4}, \frac{c+d+e+f}{4}, \frac{d+e+f+g}{4}, \ldots$

- **Method of Least Squares**

  The straight line equation, $Y = a + bX$

  Two Normal Equations, $\Sigma Y = n\,a + b\,\Sigma X$; $\Sigma XY = a\,\Sigma X + b\,\Sigma X2$

- **Methods of measuring Seasonal Variations-Method of Simple Averages:**

  Seasonal Index (S.I) = $\frac{Seasonal\ Average}{Grand\ average} X 100$

- **If the data is given in months**

  S.I for Jan = $\frac{Monthly\ Average\ (for\ Jan)}{Grand\ average} X 100$

- **If the data is given in quarter**

  S.I for Kth Quarter = $\frac{Average\ of\ K^{th}\ quarter}{Grand\ average} X 100$

- Continuous distribution function

  If X is a continuous random variable with the probability density function $f_X(x)$, then the function $F_X(x)$ is defined by

- Weighted Index Number

  Price Index (P_{01}) = $\frac{\sum p_1 w}{\sum p_0 w} x 100$

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